

Chapters 6–7 Warm-Up: Functions and Modules

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1 Starting Data

1. We will review functions and modules by extending the patient-database ideas from Problem Set 4. Suppose a patient record is represented by a list:

```
patient = [1042, "Maria Garcia", 37]
```

The entries represent the patient ID, name, and age.

2 Functions and Parameters

1. Write a function

```
def display_patient(patient):  
    # your code here
```

that prints the record in the form

```
ID: 1042  
Name: Maria Garcia  
Age: 37
```

- In the function definition, what is the **parameter**?
- When you call `display_patient(patient)` what is the **argument**?

3 Returning a Result

1. Consider the database

```
patients = [  
    [1042, "Maria Garcia", 37],  
    [1057, "Daniel Kim", 52],  
    [1081, "Aisha Patel", 29]  
]
```

Write a function

```
find_patient(patients, patient_id)
```

that searches through the database by patient ID. If the ID is found, the function should **return patient**. If no record matches, use **return None**. A useful structure is

```
def find_patient(patients, patient_id):  
    for patient in patients:  
        if patient[0] == patient_id:  
            return patient  
    return None
```

Why is **return** more useful here than simply printing the matching record inside the function?

4 Using a Module

1. Place the two functions

```
display_patient()  
find_patient()
```

inside a file called

```
patient_tools.py
```

In a second Python file or Google Colab notebook, import the module:

```
import patient_tools
```

Then search for patient ID 1057:

```
result = patient_tools.find_patient(patients, 1057)
```

Display the result using the function from the module:

```
patient_tools.display_patient(result)
```

- Which file **defines** the functions?
- Which file or notebook **calls** the functions?
- What value is stored in **result**?

5 Think Ahead to Chapter 8

1. Our current search requires an exact numeric patient ID. Suppose instead that the user remembers only part of a patient's name:

```
"Garcia"
```

What new kind of operation would the program need in order to locate the correct record?

Takeaway: Chapters 6–7 allow us to organize programs into reusable functions and modules. Chapter 8 will allow us to make the patient search more flexible by examining and matching strings.